POKHARA UNIVERSITY

Semester: Spring

Level: Bachelor
Programme: BE
Course: Microprocessors

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Differentiate between Von-Neumann and Harvard architecture. 5
b) Explain in detail the flags of 8085 microprocessor. 5
c) Write an 8085 ALP to find out the smallest number in an array available from memory location starting from 2500H to 2510H and Store the result in the register C. 5

a) What happens when HLT instruction is executed in processor? 4
b) Write an 8085 ALP to multiply the number 37H and 42H and keep the result in memory. 5

c) Design a circuit to interface a 2 KB EPROM to 8085 microprocessor .The starting address being 1000H. 6

a) Draw the timing diagrams for the instruction STA 400FH. What does quality factor mean in microprocessor? 7+1

b) What is control word? Write a program to take input from 8 switches connected to port B of 8255A and display the status of switches of 8, LED connected to port A of 8255A (for mode 0). 7

a) With the help of the register organization diagram explain the types of registers in 8086 microprocessor. 7

b) What is interrupt and interrupt service routine? Explain polled, vectored interrupt and non-maskable interrupt with examples. 8

a) Define "Assembler Directives". Explain the following assembler directives along with their format: The Memory Model Definition, The PROC directive, The .STARTUP directive, The .EXIT directive and The END directive. 7

b) Define Procedure and Macros and explain the addressing mode used in 8086 microprocessor with examples. 8

6. a) Write an 8086 ALP for MASM to display the text "POKHARAUNIVERSITY" in reverse order. 8
b) Two computers are to be connected with each other via modems. Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standards? 7

7. Write short notes on: (Any two) 2×5
a) RS 422A
b) SRAM Vs DRAM
c) 80386 microprocessor and its type

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